Developing and Operationalising Natural Language Processing solutions for Child Safeguarding: A Critical Review of Technical Possibilities and Organisational Realities in UK Policing – TITLE TO BE CONFIRMED!!!

Theoretical Perspectives of Advanced Professional Practice - MOD006047

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May 13, 2026

Abstract

Purpose: Lorem impsum...

Method: Lorem impsum...

Findings: Lorem impsum...

Conclusion: Lorem impsum...

Contents

Acronyms

AI Artificial Intelligence. 10

CSA Child Sexual Abuse. 6–8, 11, 13, 14

CSEW Crime Survey for England and Wales. 13, 14

NLP Natural Language Processing. 6–8, 10, 11, 13

NSPCC National Society for the Prevention of Cruelty to Children. 14

PKMS Personal Knowledge Management System. 9

PRISMA Preferred Reporting Items for Systematic reviews and Meta-Analyses. 10, 12

UK United Kingdom. 13

VKPP Vulnerability Knowledge & Practice Portfolio. 13

ZK Zettelkasten. 9

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Chapter 1

Introduction

1.1 Research context

This paper defines the research context before selecting the most appropriate methodology and strategy for the literature review which follows.

Empirical observations, supported by academic and grey literature, (Finkelhor et al., 2024; National Police Chiefs' Council, 2025) identify that reported instances of Child Sexual Abuse (CSA) are increasing, with resultant demand on policing outstripping proactive preventative capabilities.

The gap between demand and capability in both CSA and other operational areas of policing is acknowledged across the sector, generating political and organisational appetite for technological solutions, (National Police Chiefs' Council, 2026). Despite this appetite, evidence from both professional and academic sources identifes that policing is structurally complex and requires reform to better adopt technology, (Cooper, 2024), with cultural barriers further diminishing the successful use of technological solutions, (Thompson and Manning, 2021; Kassem and Erken, 2025).

These challenges are significant, yet they do not negate the potential of emerging technologies to address the identified capability gap. Advances in Natural Language Processing (NLP) technologies present an opportunity to deliver preventative safeguarding in the CSA context, acknowledging that the gap between technical possibility and operational deployment is not primarily technical but organisational.

This paper explores how to bridge this gap, drawing on four thematic research areas to develop a conceptual framework capable of addressing the overarching research challenge: *How can natural language processing methods be developed and operationalised to identify precursors of contact sex offending against children, to enhance proactive safeguarding opportunities within the existing operational constraints of policing?*

The four thematic research areas are as follows:

1. What linguistic precursors of contact sex offending exist in offender-victim conversations?
2. Which NLP methods can be most effectively realised within policing infrastructure?
3. What implementation barriers exist within policing that may hinder NLP adoption?
4. What strategies can be developed to overcome those barriers?

1.2 Purpose and scope of the paper

This research is inherently interdisciplinary, positioned at the intersection of technical and organisational domains. Accordingly, a literature review will require a multi-track approach, with each track addressing a different aspect of the research challenge. These aspects can be summarised as follows:

1. The first track explores the landscape of CSA and the linguistic characteristics which precede contact offending, establishing the empirical foundation for a technical solution.
2. The second track reviews the technical literature on NLP applications in safeguarding and criminal justice contexts, focusing on grooming language detection and the computational methods available for identifying linguistic precursors of contact offending.
3. The third track examines the organisational dynamics of technology adoption in policing, drawing on both academic literature and empirical evidence of structural and cultural barriers to technical innovation.
4. The fourth track addresses the ethical and governance boundaries within which any NLP solution must operate, encompassing algorithmic bias, data protection, proportionality, and the ethical context surrounding preventative safeguarding.

Each track is explored in depth, with the gathered insights being synthesised to identify where technical possibilities meet organisational realities. This structure enables a comprehensive understanding of both the capabilities and constraints shaping both the research problem and the practice improvement opportunity and grounding any solution in organisational and ethical realities

Chapter 2

Literature Search Strategy and Methodology

2.1 Review Methodology and Rationale

This literature review adopts a systematic integrative approach. Snyder (2019) documents a number of review types, noting that an integrative review seeks to combine multiple perspectives, synthesise those views, and to develop a new framework or perspective.

The interdisciplinary nature of this research, noted in section 1.2, necessitates a review methodology that can accommodate multiple perspectives and evidence types. An integrative design as defined by Torraco (2016) and Torraco (2005) allows for a review and synthesis across domains, building a conceptual framework which seeks to unify the disparate domains into a framework for operationalising and implementing an NLP solution into a safeguarding CSA context.

A systematic approach is layered into this integrative approach to allow for a targeted and transparent search and selection process in reviewing the evidence, ensuring that the review is both comprehensive and replicable. This systematic layer is particularly important given the breadth of this review, which spans technical, organisational and ethical domains, each with their own bodies of literature and evidence types. Whittemore and Knafl (2005) identifies that by adding a systematic approach to an integrative review, findings and subsequent synthesis across domains can lead to the delivery of evidence based practice improvements.

From an academic perspective, a doctoral literature review needs to be defensible and reproducible. Whilst the integrative approach allows for synthesis and framework development, a layered systematic approach ensures reproducibility whilst adding a counter balance to my own personal biases and my tendency to approach problems with a solutionist mindset.

2.2 Tools and software

The multifaceted nature of this review demands a robust literature and note management strategy supported by appropriate tools. Basu (2020) describes the Zettelkasten (ZK) method which can be used to manage literature sources and create links between them, supporting the synthesis process. This is furthered by Reck (2023) who describes the ZK method as a process of creating notes on individual literature sources, applying tags to those notes and then using the tagging system to create links between the notes. Both Basu (2020) and Reck (2023) describe the ZK method as the process of building a Personal Knowledge Management System (PKMS). Cheong and Tsui (2011) describes a PKMS as a structure which allows for the development of "*personal information management, personal knowledge internalization, personal wisdom creation, and interpersonal knowledge transferring*"

Reck (2023) considers the use of various software tools to support the ZK method. Following a period of experimentation with various applications, a software stack to support the ZK method for this review was selected consisting of Zotero for literature management and bibliography generation, Obsidian for note taking and synthesis and LaTeX for writing and formatting the review.

Zotero was selected for its robust literature management capabilities, including the ability to capture metadata, PDFs and notes on literature sources, and to auto generate and update a master bibliography file. Obsidian was selected for its flexibility in note taking and synthesis, allowing for the creation of a network of linked notes which can be easily navigated and searched. LaTeX was selected for its professional formatting capabilities, particularly for handling citations, figures and tables, and for its ability to draw directly from Zotero's auto-generated bibliography, thus ensuring consistency and ease of referencing.

All content is then automatically synced to a cloud store and defined content saved a second time to a GitHub repository. My use of the ZK method and the associated software stack is illustrated in figure 2.1.

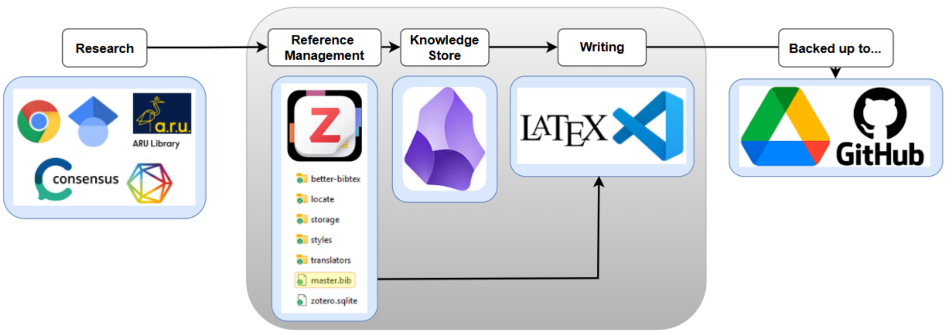

Figure 2.1: Zettelkasten method application for literature and note management

2.3 Search approach and databases

The following sections outline the search strategy for this review, including the selection of databases, keywords and search terms, and the inclusion and exclusion criteria. Section 2.6 details the returns following implementation of the search strategy and the subsequent screening and screening process which led to the final selection of literature for review. Section 1.2 identifies four distinct research tracks. Each of these tracks has a unique focus which requires a dedicated search strategy. The holistic search strategy presented in appendix A identifies the question to be answered for each track, the selected databases for each track, and the rationale for selecting those databases. These columns evidence a search strategy which is targeted and relevant to each track, and which can be replicated by other researchers. The search strategy is also designed to ensure that the resulting literature return is manageable, given the breadth of the research challenge and the interdisciplinary nature of the review.

2.4 Keywords and search terms

Key words and search terms are critical to the success of a literature review. Chigbu et al. (2023) provides guidance on keyword selection and submission to selected databases. The keywords and search terms utilised for each of the noted research tracks are defined in appendix B. These align to the search strategy outlined in appendix A. The keywords and search terms are designed to capture thematic content relevant to each track. Boolean operators (AND, OR, NOT) are used to combine keywords and search terms in a way that maximises the relevance of retrieved literature.

2.5 Inclusion and exclusion criteria

Appendix C defines the inclusion and exclusion criteria for each of the four research tracks. These criteria are designed to ensure that the literature included in the review is relevant, high quality, and aligned with the research challenge. The year date of 2018 is presented as this is largely considered to be the breakout year for the devlopment of NLP transformer based models, (Khanna, 2024), a technical architecture which will be relevant to this research.

2.6 Results and filtering process

A Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) diagram is presented in figure 2.3 which summarises the selection process following the implementation of the search strategy.

An initial search return of 3,150 papers was obtained. This initial return was reduced to 1,471 following the removal of duplicate papers, foreign language papers and papers with no immediately viewable author.

These 1,471 were subject to a first pass screen at title level; this revealed a high degree of topic irrelevancy, potentially due to the contemporary explosion of interest in Artificial Intelligence (AI) and NLP which returned a large number of papers with excessively expansive key word flags. This titular screen removed 1,020 papers leaving 451 papers.

These 451 papers were then subject to a secondary screening at abstract level, with each paper being reviewed against the inclusion and exclusion criteria defined in appendix C. Reflecting on the inclusion and exclusion criteria during this process led to the realisation that the defined criteria were not entirely appropriate to the screening task. In some cases, the criteria were too specific, resulting in some papers which had relevant content being excluded. In other cases, the criteria did not match the content of the paper and therefore did not allow for defined inclusion or exclusion criteria. This led to a process of adherence to the criteria where possible but also implementing a deree of subjective judement where the criteria were not entirely fit for purpose. This process was supported by a note taking process which captured the rationale for inclusion or exclusion of each paper, ensuring that the process was transparent and reproducible. An example of this note taking is provided in figure 2.2.

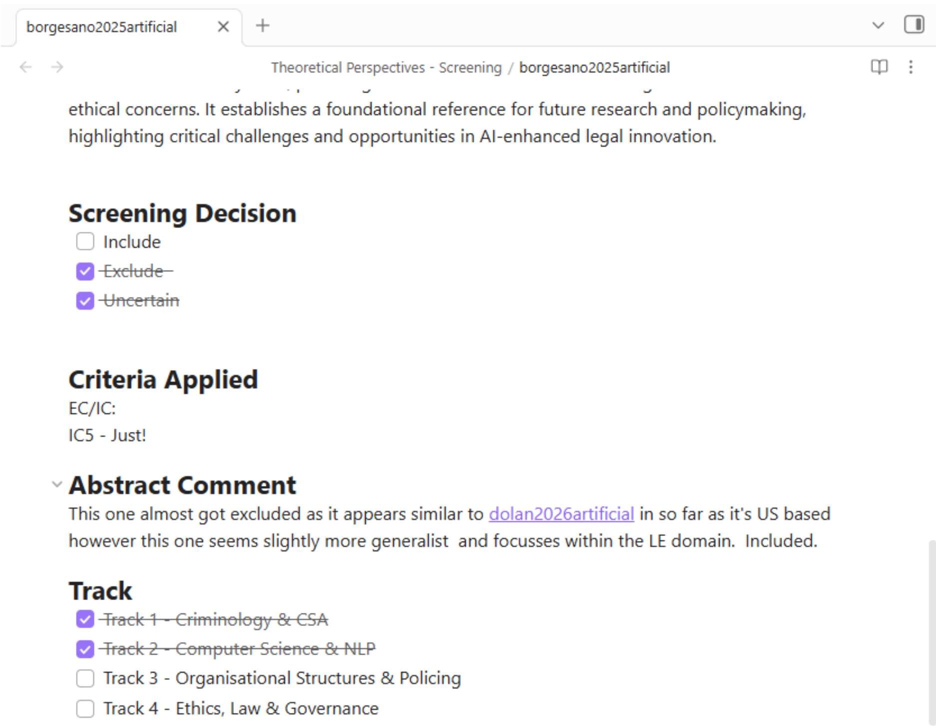

Figure 2.2: Example of exclusion criteria not being entirely fit for purpose, leading to the need for subjective judgement in the screening process

This process reduced the number of papers to 196. These papers were then subject to a tertiary screen of the abstract and conclusion with a focus on relevance specifically to either a CSA, a NLP or a UK policing context. Additionally a number of known seminal papers which had been identified through professional engagements and previous academic work were added to the final selection. A purposful selection process such as this is noted by Rethlefsen et al. (2021) as a technique to ensure that known seminal works not captured by the database searches (for example those excluded by time periods) are included in the review.

This completed screening process resulted in a final selection of 94 papers for review. The selection process is summarised in the PRISMA diagram presented in figure 2.3.

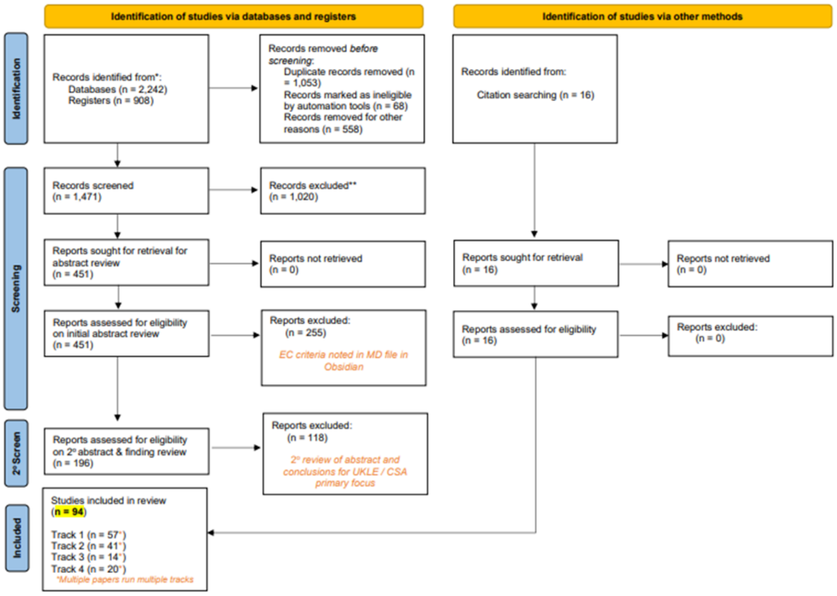

Figure 2.3: PRISMA diagram summarising the literature search and filtering process

Chapter 3

The Landscape of Child Sexual Exploitation and Policing's Response

This chapter explores the scale of CSA reporting and the resultant demand on law enforcement. This chapter also explores the grooming to contact offending pathway. Understanding this pathway is critical to the development of an NLP solution which seeks to identify precursors of contact offending, as it provides the empirical foundation for identifying the linguistic characteristics which may be present in offender-victim conversations prior to contact offending.

3.1 The scale of CSA reporting

The measurement of crime statistics in the United Kingdom (UK) is complex, with two primary sources of data providing different perspectives on crime (Ariel and Bland, 2019). The Crime Survey for England and Wales (CSEW) acts as a household survey capturing participants' self-reported experiences of crime, whilst police recorded crime data captures the volume of offences formally reported to and recorded by police forces.

Neither dataset can be considered a definitive measure of crime, and this limitation is thrown into sharp relief when examining CSA.

The Vulnerability Knowledge & Practice Portfolio (VKPP) reported 122,768 CSA offences in 2024 (VKPP, 2025) taken from police recorded crime data, whilst the 2024 CSEW estimates that 1 million people had experienced sexual abuse in the previous year, (Office for National Statistics, n.d.).

It is important to note that these two datasets are not directly comparable. The VKPP CSA figure represents criminal offences reported to police forces within a single calendar year. The CSEW figure is a cumulative prevalence estimate drawn from a cross-sectional household survey of adults aged 16 and over recalling experiences.

These discrepancies in reporting the scale of CSA offending highlights the challenges in measuring the scale of the issue and the demand it places on policing. The VKPP figure represents the formal demand on policing resources, while the CSEW figure suggests a much larger underlying prevalence of CSA that may not be fully captured by police recorded crime data. Additionally, Whilst the described reporting of CSA focusses on physical offending, Finkelhor et al. (2024) suggests that online sexual abuse and communication, when taken into consideration, would markedly increase the prevalence of CSA.

This is a view supported by Wefers et al. (2024) who leans into figures provided by the National Society for the Prevention of Cruelty to Children (NSPCC). Whilst the source of Wefers et al. (2024)'s NSPCC data is no longer accessible, more contempory reporting by the NSPCC suggests an 89% increase in sexual communication offences between 2018 to 2024, (NSPCC, 2024).

Starkly, these NSPCC figures relate to only 7,000 reports captured by NSPCC Freedom of Information requests to police forces, (NSPCC, 2024). The research of Finkelhor et al. (2024) and Wefers et al. (2024), when aligned to this low number, suggests that the scale of CSA associated to online sexual abuse and communication is not being fully captured by police recorded crime data, and that there may be a significant gap between the prevalence of online sexual communications and the demand it places on policing resources.

This gap suggests that the potential demand on policing resources may be much higher than what is formally reported, and that there may be a large number of unreported cases of CSA associated with online sexual communication. This has implications for the development of technological solutions to address the gap between prevalence and reporting and enhancing proactive safeguarding opportunities within the existing operational constraints of policing.

3.2 The demand on law enforcement

Both academic and grey literature identify that the demand on policing resources from CSA is increasing, with the gap between demand and capacity widening.

Jay et al. (2022) suggests that the scale of known CSA demand may never be met by law enforcement, identifying that digital forensic delays and a constantly shifting set of policing priorities both impact on law enforcement's ability to tackle the issue. Casey (2025) adds to this summation, noting complexities around fragmented polcing systems and a lack of intelligence sharing further exacerbate the potential to tackle demand.

Whilst these comments focus on the known demand, Winters et al. (2022) suggests that law enforcement may never have the capacity to proactively detect the grooming behaviours which may lead to the commission of sexual offences. This is a point which is reinforced by Li (2020) who describes the lengthy manual processes involved in identifying such behaviours and then targetting suspects in a proactive manner.

The described grooming behaviours, and the complexities described all reflect the reality of the comments of Finkelhor et al. (2024) and the findings of the CSEW, that the lack of data concerning online sexual communications and abuse masks the scale of the CSA issue.

3.3 The grooming to contact offending pathway

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3.4 The case for technological intervention - Identifying the research opportunity

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Chapter 4

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8.2 How the literature has shaped the framework

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8.3 Refined research aim and sub-questions

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9.1 Summary of the argument

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9.2 Implications for professional practice

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9.3 Next steps

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Word count (Chapters 1–N): XXX words, calculated in accordance with standard academic convention, excluding references, figures, tables, captions, running headers, and front matter.

Appendix A

Search Matrix

Analytical Question	Databases	Database Justification
Track 1 — Criminology		
What linguistic and behavioural precursors of contact sex offending exist in offender–victim conversations, and how do grooming typologies characterise the progression toward contact abuse?	• Scopus • PsycINFO • Web of Science • Criminal Justice Abstracts • ProQuest • Google Scholar	• **Scopus** — Broad peer reviewed coverage across policing and criminology topics. • **PsycINFO** — Consideration of psychological and behavioural dimensions of offending. • **Web of Science** — Broad peer reviewed coverage across policing and criminology topics. • **Criminal Justice Abstracts** — Dedicated criminological literature inclusive of coverage of sex-offending typologies. • **ProQuest** — Includes the Criminal Justice Database focussed on crime-related research, law enforcement practices, and police administration. • **Google Scholar** — Grey literature, NGO reports, and regulatory publications alongside mainstream citation trails.
Track 2 — Computer Science & NLP		
Analytical Question	**Databases**	**Database Justification**
Which NLP methods can be most effectively realised within policing infrastructure, and what does the evidence base reveal about model performance, explainability, and operational deployment constraints?	• IEEE Xplore • ACM Digital Library • ACL Anthology • arXiv (cs.CL, cs.AI, cs.CY) • Papers With Code • Scopus	• **IEEE Xplore** — Institute of Electrical and Electronics Engineers (IEEE) curated database for applied NLP architecture, development and deployment. • **ACM Digital Library** — Curated by the Association for Computing Machinery (ACM); similar in stature for NLP content as **IEEE Xplore**. • **ACL Anthology** — Curated by the Association for Computational Linguistics (ACL); linguistics-grounded NLP resource with a focus on language understanding and generation, essential for conversational analysis. • **arXiv** (cs.CL) — Pre-print access to novel language model research. • **Papers With Code** — Community driven repository linking code implementations to research papers, crucial for identifying reproducible NLP methods and benchmarks. • **Scopus** — Interdisciplinary bridge between technical and applied criminological literature.

Analytical Question	Databases	Database Justification
Track 3 — Organisational Structures & Policing		
What structural, cultural, and institutional factors act as barriers to or enablers of NLP adoption in policing organisations, and how does change management theory account for technology resistance in law enforcement?	• Web of Science • Criminal Justice Abstracts • PsycINFO • ProQuest • Dimensions • Google Scholar	• **Web of Science** — Peer-reviewed journals across the social sciences, inclusive of law enforcement and organisational change. • **Criminal Justice Abstracts** — Focused research on policing and policing culture. • **PsycINFO** — Human and organisational literature featuring law enforcement contexts. • **ProQuest** — Includes the Criminal Justice Database focussed on crime-related research, law enforcement practices, and police administration. • **Dimensions** — Artificial Intelligence (AI) powered search tool for the identification of emerging research clusters in public-sector innovation. • **Google Scholar** — Established open source search engine, used here for grey literature and policy reports from law-enforcement agencies and inspectorates.

Analytical Question	Databases	Database Justification
Track 4 — Ethics, Law & Governance		
What ethical and regulatory constraints govern NLP deployment in law enforcement, and how should proportionality, data protection, and rights-based frameworks shape the design and oversight of such systems?	• HeinOnline • SSRN • EUR-Lex • PhilPapers • ProQuest & Scopus • Dimensions	• **HeinOnline** — Links to the Law Journal Library containing peer reviewed articles on the ethical implications of AI bias. • **SSRN** — Social Science Research Network (SSRN) contains dedicated sections on criminal justice research focussed on AI and technology adoption. • **EUR-Lex** — Access to European Union (EU) legislative instruments, including the AI Act and GDPR, relevant to algorithmic oversight in law enforcement across the EU. • **PhilPapers** — Repository of academic literature exploring the intersection of ethics, law, and governance regarding police use of AI. • **ProQuest & Scopus** — Both offer broad peer reviewed coverage across policing and criminology topics with crossover in ethical and legal research. • **Dimensions** — AI powered search tool for the identification of emerging research clusters in ethical and legal dimensions of AI deployment.

Appendix B

Search Terms and Boolean Operators

Terms	Boolean Strings
Track 1 — Criminology	

• Online grooming
• Sexual grooming
• Grooming behaviour
• Child sexual abuse
• CSA
• CSAM
• Grooming language
• Grooming communication
• Child
• Victim
• Sex offend*
• Abuse
• Contact offending
	`("online grooming" OR "sexual grooming" OR "grooming behaviour") AND ("child sexual abuse" OR "CSA" OR "CSAM")`
	
	`("grooming language" OR "grooming communication") AND ("child" OR "victim") AND ("sex offend*" OR "abuse")`
	
	Proximity (Scopus / WoS):
	`"grooming" NEAR/5 "contact offending"`

Track 2 — Computer Science & NLP	

• BERT
• RoBERTa
• DistilBERT
• Text classification
• Sequence classification
• Grooming
• Predatory
• Child safety
• Explainable AI
• XAI
• Interpretable machine learning
• NLP
• Natural language processing
• Law enforcement
• Policing
• Criminal justice
• PAN shared dataset
• Perverted Justice
• Chat log
• Online conversation
• Grooming detection
• Predatory communication
• Sexual solicitation
	`("BERT" OR "RoBERTa" OR "DistilBERT") AND ("text classification" OR "sequence classification") AND ("grooming" OR "predatory" OR "child safety")`
	
	`("explainable AI" OR "XAI" OR "interpretable machine learning") AND ("NLP" OR "natural language processing") AND ("law enforcement" OR "policing" OR "criminal justice")`
	
	`("PAN shared dataset" OR "Perverted Justice" OR "chat log" OR "online conversation") AND ("grooming detection" OR "predatory communication" OR "sexual solicitation")`

Track 3 — Organisational Studies & Policing	
Terms	**Boolean Strings**

• Technology adoption
• Technology resistance
• Organisational inertia
• Policing
• Law enforcement
• Police force
• Tacit knowledge
• Explicit knowledge
• Knowledge management
• Police officer*
• Police culture
• Innovation
	`("technology adoption" OR "technology resistance" OR "organisational inertia") AND ("policing" OR "law enforcement" OR "police force")`
	
	`("tacit knowledge" OR "explicit knowledge" OR "knowledge management") AND ("policing" OR "police officer*" OR "law enforcement")`
	
	Proximity (WoS):
	`"police culture" NEAR/8 ("technology resistance" OR "innovation")`

Track 4 — Ethics, Law & Governance	

• Algorithmic bias
• Automated decision-making
• Algorithmic accountability
• Law enforcement
• Policing
• Criminal justice
• Ethics
• Human rights
• Fairness
• Data protection
• GDPR
• UK DPA
• NLP
• Machine learning
• Artificial intelligence
	`("algorithmic bias" OR "automated decision-making" OR "algorithmic accountability") AND ("law enforcement" OR "policing" OR "criminal justice") AND ("ethics" OR "human rights" OR "fairness")`
	
	`("data protection" OR "GDPR" OR "UK DPA") AND ("NLP" OR "machine learning" OR "artificial intelligence") AND ("policing" OR "criminal justice")`

Appendix C

Inclusion and Exclusion Criteria

Inclusion Criteria

- **IC1** Peer-reviewed articles, conference proceedings, and edited book chapters, 2018–present.
- **IC2** Grey literature from authoritative sources (National Police Chiefs' Council (NPCC), Home Office, Child Exploitation and Online Protection Centre (CEOP), National Institute of Standards and Technology (NIST), Information Commissioner's Office (ICO)), 2018–present.
- **IC3** NLP, machine learning, or computational linguistics applied to safeguarding, criminal justice, or analogous operational contexts.
- **IC4** Grooming, CSA, or online predatory behaviour with linguistic, behavioural, or empirical focus.
- **IC5** Organisational, cultural, or governance studies on technology adoption in policing or structurally comparable public-sector contexts.
- **IC6** English-language sources; non-English permitted where a peer-reviewed translation or substantive abstract provides sufficient evidential grounding with explicit methodological justification.
- **IC7** Pre-2018 foundational works where theoretical significance is established and no updated equivalent exists, subject to explicit justification in the synthesis narrative.

Exclusion Criteria

- **EC1** Sources prior to 2018 unless meeting IC7; all exceptions recorded in the review log.
- **EC2** Non-peer-reviewed opinion pieces, blog posts, or media articles without evidential grounding, unless from authoritative professional or statutory bodies.
- **EC3** Studies focused solely on adult sexual offending with no transferable relevance to child safeguarding contexts.
- **EC4** Technical NLP literature without contextual application to safeguarding, criminal justice, or operational deployment (foundational instructional / code text not included)
- **EC5** Technology adoption studies from sectors lacking structural or cultural comparability to UK policing (e.g. private-sector contexts without analogous accountability frameworks).
- **EC6** Duplicate publications; where multiple versions exist, the most recent peer-reviewed version is retained.
- **EC7** Sources for which full text cannot be obtained via institutional access, inter-library loan, or open-access repositories within a reasonable retrieval window.

Appendix D

Patch One - Upload and Feedback

Chapter 1

Introduction

1.1 Research context

This paper defines the research context before selecting the most appropriate methodology and strategy for the literature review which follows.

Empirical observations, supported by academic and grey literature, (Finkelhor et al., 2024; National Police Chiefs' Council, 2025) identify that reported instances of Child Sexual Abuse (CSA) are increasing, with resultant demand on policing outstripping proactive preventative capabilities.

The gap between demand and capability in both CSA and other operational areas of policing is acknowledged across the sector, generating political and organisational appetite for technological solutions, (National Police Chiefs' Council, 2026). Despite this appetite, evidence from both professional and academic sources identifes that policing is structurally complex and requires reform to better adopt technology, (Cooper, 2024), with cultural barriers further diminishing the successful use of technological solutions, (Thompson and Manning, 2021; Kassem and Erken, 2025).

These challenges are significant, yet they do not negate the potential of emerging technologies to address the identified capability gap. Advances in Natural Language Processing (NLP) technologies present an opportunity to deliver preventative safeguarding in the CSA context, acknowledging that the gap between technical possibility and operational deployment is not primarily technical but organisational.

This paper explores how to bridge this gap, drawing on four literature tracks to develop a conceptual framework capable of addressing the proposed research challenge: *How can natural language processing methods be developed and operationalised to identify precursors of contact sex offending against children, to enhance proactive safeguarding opportunities within the existing operational constraints of policing?*

This challenge consists of four thematic research areas:

1. What linguistic precursors of contact sex offending exist in offender-victim conversations?
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1.2 Purpose and scope of the paper

This research is inherently interdisciplinary, positioned at the intersection of technical and organisational domains. Accordingly, a literature review will require a multi-track approach, with each track addressing a different aspect of the research challenge. The four tracks are as follows:

1. The first track explores the landscape of CSA and the linguistic characteristics which precede contact offending, establishing the empirical foundation for a technical solution.
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A systematic approach is layered into this integrative approach to allow for a targeted and transparent search and selection process in reviewing the evidence, ensuring that the review is both comprehensive and replicable. This systematic layer is particularly important given the breadth of this review, which spans technical, organisational and ethical domains, each with their own bodies of literature and evidence types. Whittemore and Knafl (2005) identifies that by adding a systematic approach to an integrative review, findings and subsequent synthesis across domains can lead to the delivery of evidence based practice improvements.

From an academic perspective, a doctoral literature review needs to be defensible and reproducible. Whilst the integrative approach allows for synthesis and framework development, a layered systematic approach ensures reproducibility whilst adding a counter balance to my own personal biases and my tendency to approach problems with a solutionist mindset.

2.2 Tools and software

The multifaceted nature of this review demands a robust literature and note management strategy supported by appropriate tools. Basu (2020) describes the Zettelkasten (ZK) method which can be used to manage literature sources and create links between them, supporting the synthesis process. This is furthered by Reck (2023) who describes the ZK method as a process of creating notes on individual literature sources, applying tags to those notes and then using the tagging system to create links between the notes. Both Basu (2020) and Reck (2023) describe the ZK method as the process of building a Personal Knowledge Management System (PKMS). Cheong and Tsui (2011) describes a PKMS as a structure which allows for the development of "*personal information management, personal knowledge internalization, personal wisdom creation, and interpersonal knowledge transferring*"

Reck (2023) considers the use of various software tools to support the ZK method. Following a period of experimentation with various applications, a software stack to support the ZK method for this review was selected consisting of Zotero for literature management and bibliography generation, Obsidian for note taking and synthesis and LaTeX for writing and formatting the review.

Zotero was selected for its robust literature management capabilities, including the ability to capture metadata, PDFs and notes on literature sources, and to auto generate and update a master bibliography file. Obsidian was selected for its flexibility in note taking and synthesis, allowing for the creation of a network of linked notes which can be easily navigated and searched. LaTeX was selected for its professional formatting capabilities, particularly for handling citations, figures and tables, and for its ability to draw directly from Zotero's auto-generated bibliography, thus ensuring consistency and ease of referencing.

All content is then automatically synced to a cloud store and defined content saved a second time to a GitHub repository. My use of the ZK method and the associated software stack is illustrated in figure 2.1.

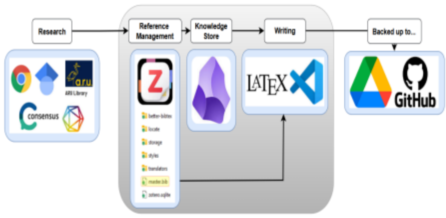

Figure 2.1: Zettelkasten method application for literature and note management

2.3 Search approach and databases

Section 1.2 identifies four distinct research tracks. Each of these tracks has a unique focus and therefore requires a dedicated search strategy to maximise the relevance and quality of included literature. The search strategy presented in table ?? defines the four core research tracks, the question to be answered for each track, the selected databases for each track, and the rationale for selecting those databases. The search strategy is designed to ensure that the literature review is comprehensive, targeted and relevant to the research challenge and that it can be replicated by other researchers. The search strategy is also designed to ensure that the literature review is manageable, given the breadth of the research challenge and the interdisciplinary nature of the review.

2.4 Keywords and search terms

Key words and search terms are critical to the success of a literature review. Chigbu et al. (2023) provides holistic guidance on keyword selection and submission to selected databases. The keywords and search terms utilised for each of the noted research tracks are defined in table ??. These align to the search strategy outlined in table ??. The keywords and search terms are designed to capture the key concepts and themes relevant to each track, ensuring that the literature retrieved is relevant to the research challenge.

aru

Patrick Thompson

18 Mar 20:18 | Last reply 3 Apr 23:42

Hi squad!

I've uploaded a PDF entitled "Draft so far..." which is an effort to get some structure around my literature review.

I've not made much headway on actually reading lots yet as I wanted to nail down the strategy.

Hopefully the document makes logical sense. Essentially I've broken down my literature review into four tracks / topics and then decided on a systematic integrative review methodology given the four slightly separate but linked topic areas.

In the files section (https://canvas.anglia.ac.uk/groups/44479/files) I've also uploaded four tables;

One is called search_matrix. Its a table were I try to justify search sources for the four tracks.

One is called keywords. It's my attempt to narrow down keywords across those four tracks.

The other two are inclusion / exclusion criteria. The 2018 date is linked to the year when large language models became more openly available and made a big difference in the world of natural lanuage processsing.

Speak soon!

Pat

Name	Created	Last Modified	Modified by	Size	Actions
exclusion_criteria.html	18 Mar 2026	18 Mar 2026	Patrick Thompson	2 KB	⋮
inclusion_criteria.html	18 Mar 2026	18 Mar 2026	Patrick Thompson	2 KB	⋮
keywords_v2.html	25 Mar 2026	25 Mar 2026	Patrick Thompson	17 KB	⋮
search_matrix.html	18 Mar 2026	18 Mar 2026	Patrick Thompson	8 KB	⋮

Beth Brown

29 Mar 14:57

Hi Pat,

For an early first draft I thought this was awesome! Your writing is clear and you articulate the research context with precision, I felt that your opening section clearly situates the problem within policing, policy, and technological landscapes, and you explain this with a balanced, evidence-based tone.

The four research tracks are logical and well reasoned, clearly connecting to your overarching question. Your tables are also incredibly clear and organised. I think the fact you have got such a clear structure will make your literature review manageable and robust. It also shows that you've thought deeply about the complexity of the problem space.

The justification for separate strategies is sound and aligns well with the assignment brief. You could consider briefly mentioning in the introduction what is out of scope and why. This may help manage expectations and demonstrate deliberate boundary setting. Also, you've clearly explained the tracks individually but a short statement about how they will be brought together (e.g., thematic synthesis, conceptual mapping, cross-domain triangulation) would provide further clarity about the approach.

I really liked your explanation of why an integrative review is appropriate, and how you're layering systematic elements to ensure rigour. This demonstrates a strong understanding of review methodologies and defended your choices well.

Overall I thought this was great - you show really clear thinking and a strong grasp of both the technical and organisational sides of the topic! Look forward to the next Patch..

Reply | Like | Mark as unread

Lauren Hunt

3 Apr 23:42

Hi Pat,

Its reassuring we're at a similar stage, but its clear you know your topic and that you've spent a lot of time thinking of the best way to tackle it!

Love the tables in the files and numbering your in/exclusion criteria to refer to. You (and Beth) have showed your Boolean search words very well and something I'd develop in my work - I also like how you clearly showed each element and then showed the cross of the topics (even if I don't understand some of the AI bits!). Its good to see how all of your tracks meet up and relate to your question - you've done well to get into the detail of the topic but also explain each section clearly.

The search matrix took me a second to get my head around, but this only helped me as again you and Beth have used many more databases/journals than me so maybe I need to explore this more to maximise results. I was worried about over-complicating my search but you've both presented this in clear formats so I may give it another go! I can also see how your feedback to me to clear in your own work regarding in/exclusion of paper dates, with 2018 being an important time for your topic.

You justify your choices well with citations and your writing flows nicely - overall I think you've made a fantastic start and time well spent setting up a great foundation - I look forward to draft 2!

Reply | Like | Mark as unread

Patrick Thompson

24 Mar 16:11

Hi Beth,

Here are my thoughts!

The keyword choices are well considered and the reflective bit around spotting "constabulary" in the results shows non-linear thinking... you're engaging with the process rather than just running through it mechanically. The Boolean searches you settled on clearly gave you a really solid foundation of papers to work from.

My dodgy maths puts you somewhere around 5,700 returns once you focus down to policing — though with three databases there's inevitably going to be a hefty overlap, so I'm guessing the real number is probably closer to 2K.... is there scope to be more robust on the exclusion criteria? You've only got one at the moment and I wonder if tightening this would actually give you more freedom, letting you really focus on contemporary practice and lived officer experience rather than having to wade through how things have evolved over the last decade and a half?

I guess along side that is the 2010 date. Sixteen years is a long stretch and I wonder if it's worth either shortening the time span or giving that choice a really solid justification, especially since the five-year Google Scholar search felt a little inconsistent alongside it.

I really liked the thematic buckets in the screening section — they're well chosen and make sense. It did make me wonder though how you're moving from the initial screening to identifying *key* papers specifically? It might be worth a sentence on what's elevating one paper above another?

The critical summaries are cool!

Reply | Like | Mark as unread

Patrick Thompson

24 Mar 16:38

Hi Lauren,

I really like the way you've set this out – the PICO structure to define the requirements is really logical, and the dual-track in-vivo and in-vitro approach makes the methodology robust.

The inclusion and exclusion criteria are sensible and clearly reasoned. One question worth considering: does there need to be any date parameter in the search, or at least an explicit justification for why all dates are included? Reviewers may ask about this.

The planned PRISMA flow chart is a great addition and will keep the whole process transparent and reproducible. Looking forward to seeing the work develop – you seem to be at a similar stage to me in having nailed your methodology and are about to start the actual capture of content.

Reply | 1 Like | Mark as unread

Appendix E

Patch Two - Upload and Feedback

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